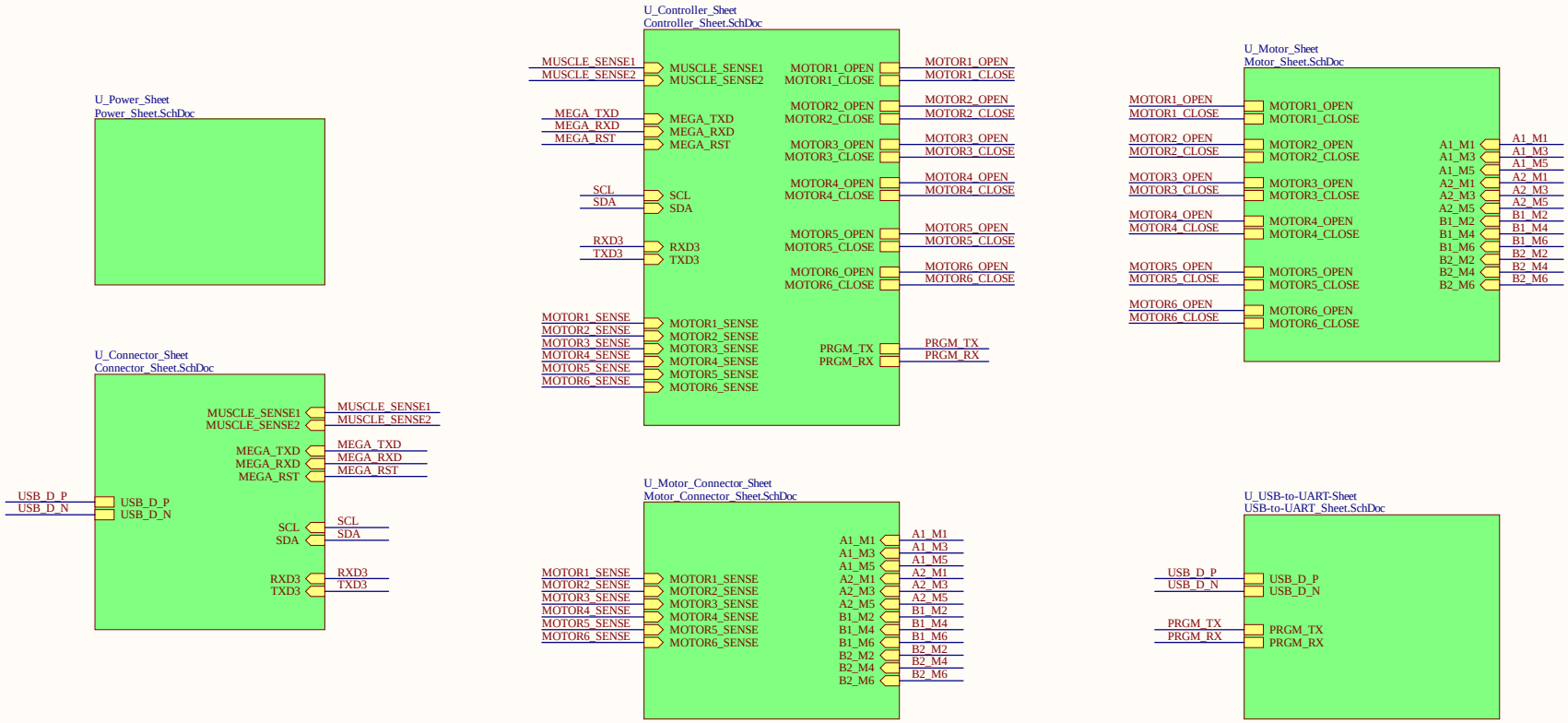
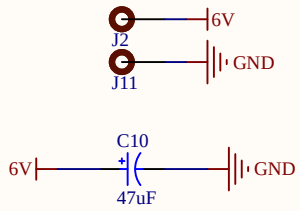
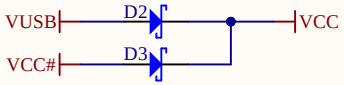

U_Power_Sheet
Power_Sheet.SchDoc
U_Connector_Sheet
Connector_Sheet.SchDoc
MUSCLE_SENSE1
MUSCLE_SENSE2
MEGA_TXD
MEGA_RXD
MEGA_RST
USB_D_P
USB_D_N
SCL
SDA
RXD3
TXD3
U_Controller_Sheet
Controller_Sheet.SchDoc
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
PRGM_TX
PRGM_RX
U_Motor_Sheet
Motor_Sheet.SchDoc
A1_M1
A1_M3
A1_M5
A2_M1
A2_M3
A2_M5
B1_M2
B1_M4
B1_M6
B2_M2
B2_M4
B2_M6
U_Motor_Connector_Sheet
Motor_Connector_Sheet.SchDoc
U_USB-to-UART-Sheet
USB-to-UART_Sheet.SchDoc

USB_D_N
USB_D_P

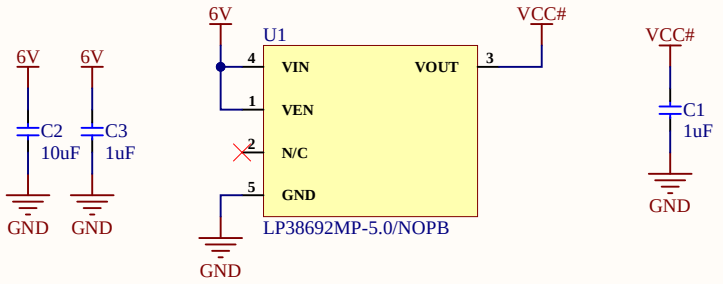
POWER INPUTS

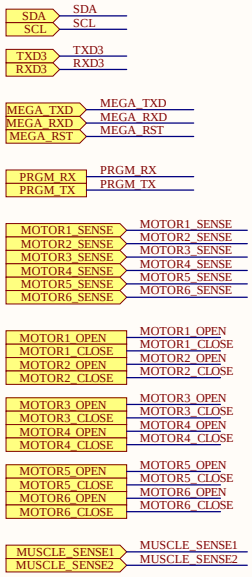


DIODE ORING

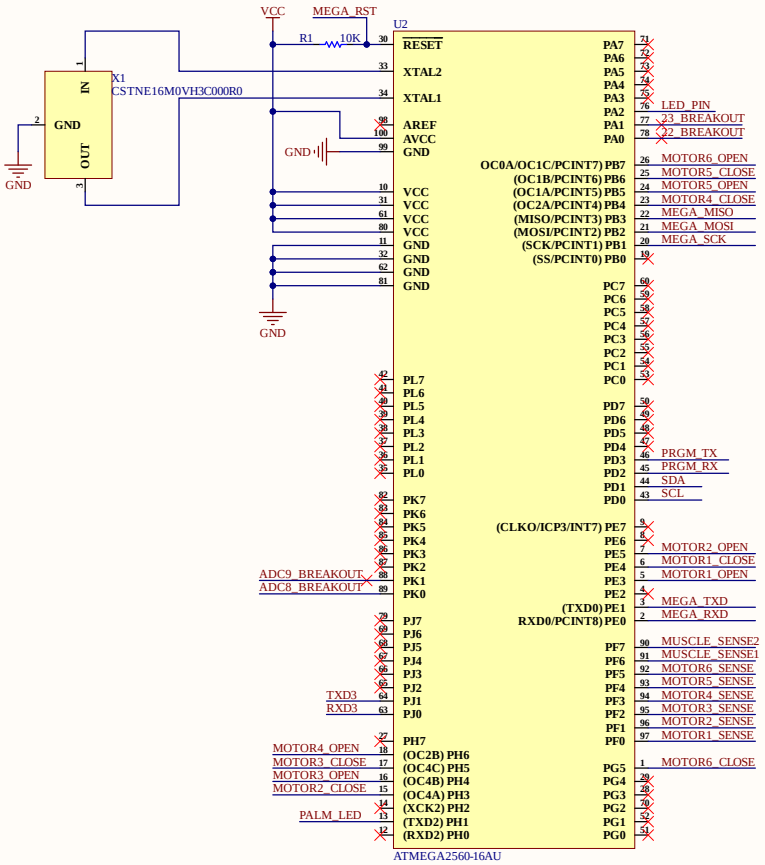


6V TO 5V CONVETER

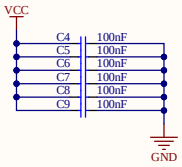




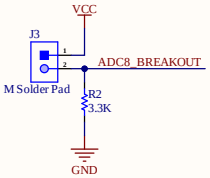
MAIN CONTROLLER



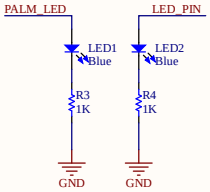
DECOUPLING CAPACITORS



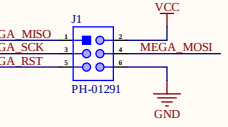
SENSOR CONNECTION



LED'S



SPI



B2_M6
B1_M6
B2_M6
B1_M6

A2_M5
A1_M5
A2_M5
A1_M5

B2_M4
B1_M4
B2_M4
B1_M4

A2_M3
A1_M3
A2_M3
A1_M3

B2_M2
B1_M2
B2_M2
B1_M2

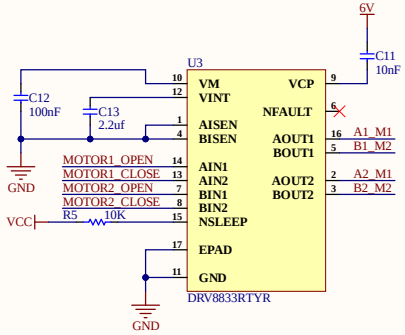
A2_M1
A1_M1
A2_M1
A1_M1

MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE

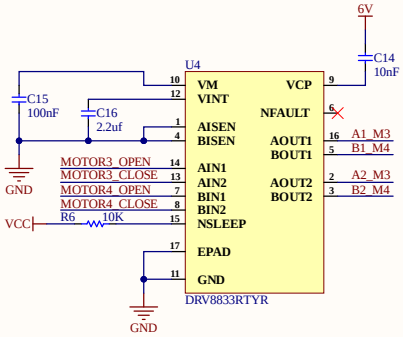
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE

MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE

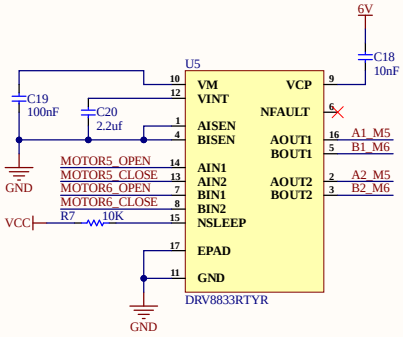
MOTOR 1 & 2

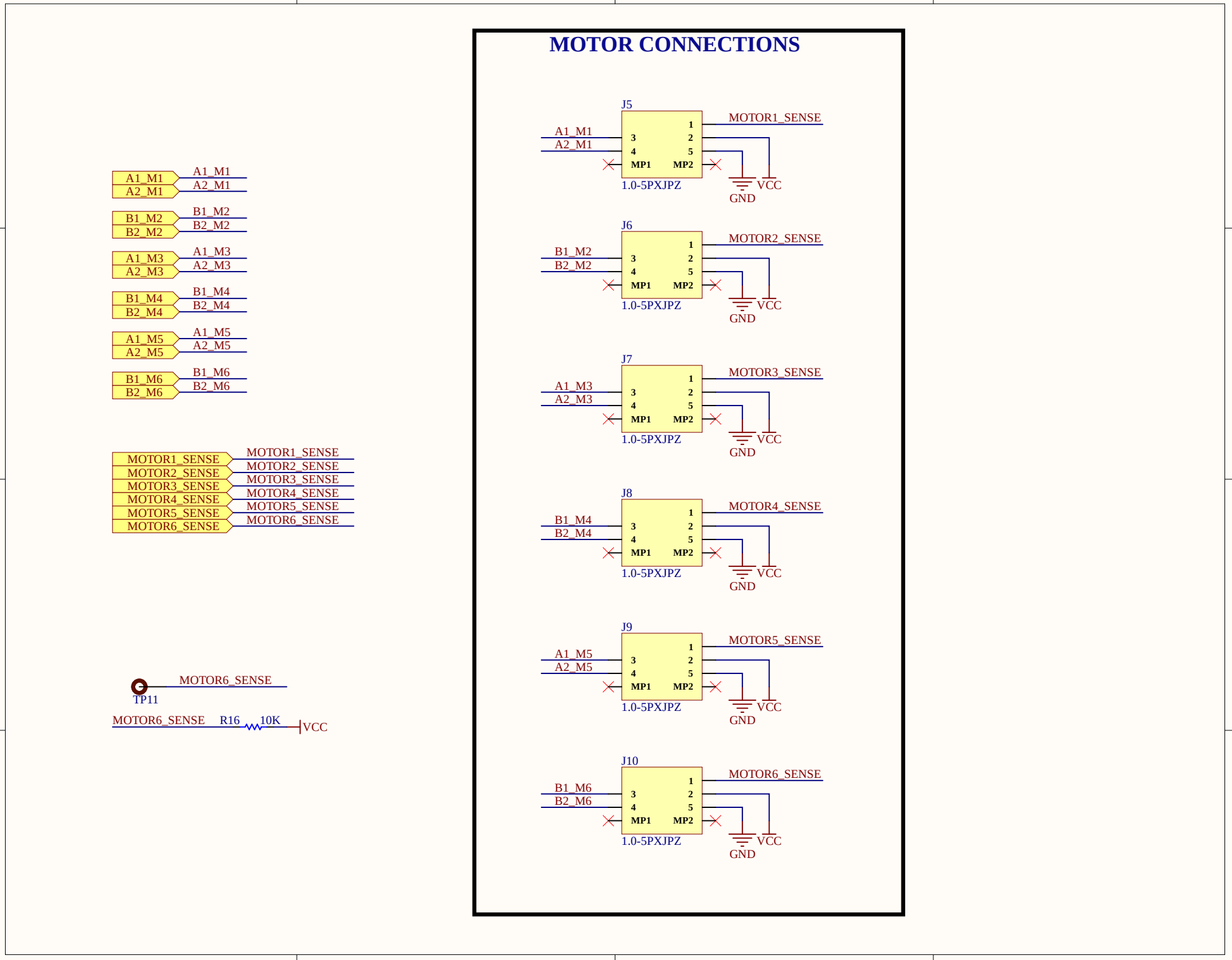


MOTOR 3 & 4

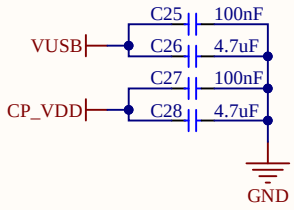


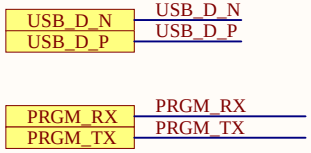
MOTOR 5 & 6

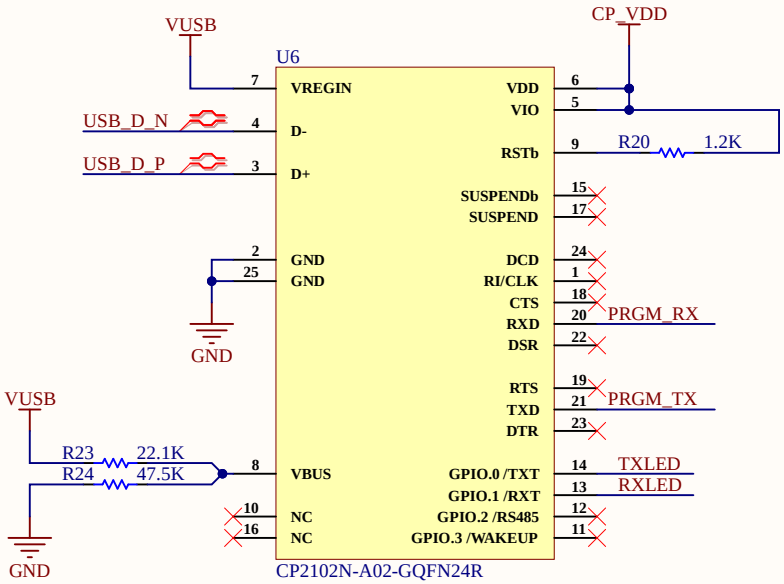


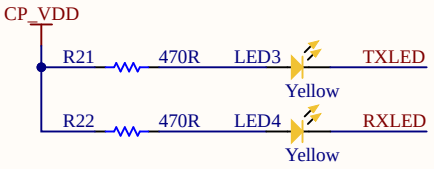

MOTOR CONNECTIONS
A1_M1
A2_M1
B1_M2
B2_M2
A1_M3
A2_M3
B1_M4
B2_M4
A1_M5
A2_M5
B1_M6
B2_M6
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
TP11
R16
10K
VCC
GND
J5
J6
J7
J8
J9
J10
MP1
MP2
1.0-5PXJPZ

USB-to-UART for PROGRAMMING

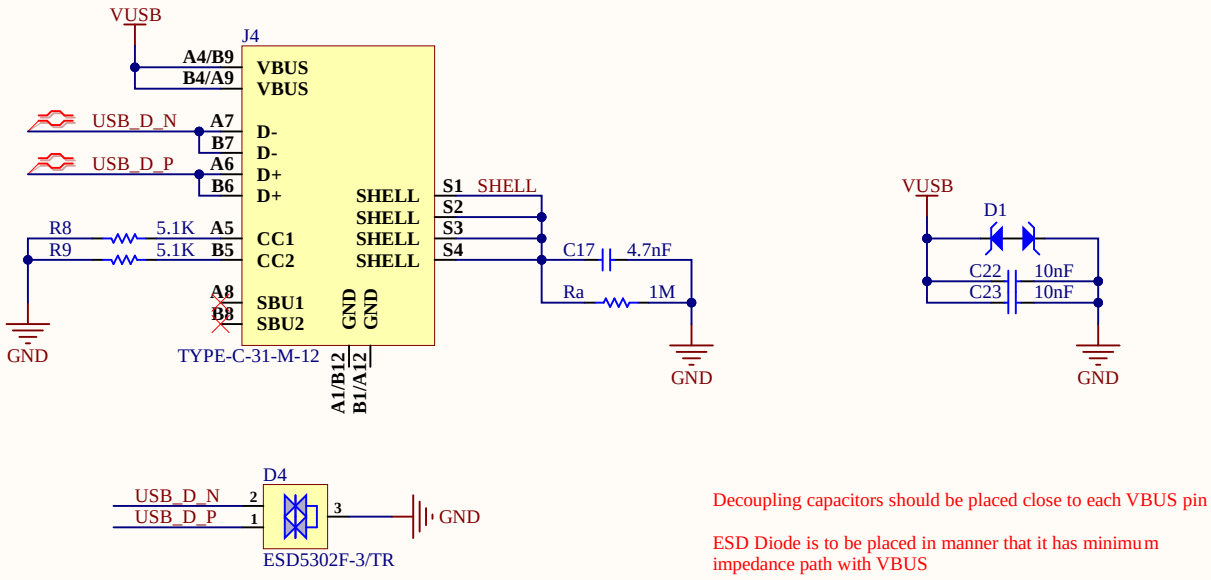








USB-C CONNECTOR



Decoupling capacitors should be placed close to each VBUS pin

ESD Diode is to be placed in manner that it has minimum impedance path with VBUS

SDA
SCL

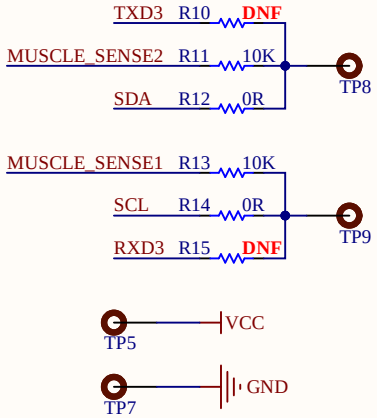
TXD3
RXD3

USB_D_N
USB_D_P

MEGA_TXD
MEGA_RXD
MEGA_RST

MUSCLE_SENSE1
MUSCLE_SENSE2

AUDIO CONNECTION



USB CONNECTION

